

Q1.

Create database

```
CREATE DATABASE DBMS_PRACTICAL;  
USE DBMS_PRACTICAL;
```

Create COURSE table

```
CREATE TABLE COURSE (  
    Course_ID INT PRIMARY KEY,  
    Course_Name VARCHAR(50),  
    Fees DECIMAL(10,2)  
);
```

Create STUDENT table

```
CREATE TABLE STUDENT (  
    Student_ID INT PRIMARY KEY,  
    Student_Name VARCHAR(50),  
    Course_ID INT,  
    Marks INT  
);
```

Insert COURSE records

```
INSERT INTO COURSE VALUES  
(101, 'BCA', 50000),  
(102, 'BSc IT', 45000),  
(103, 'BBA', 40000);
```

Insert Student records

```
INSERT INTO STUDENT VALUES  
(1, 'Rahul', 101, 85),  
(2, 'Priya', 102, 72),  
(3, 'Amit', 101, 65),  
(4, 'Sneha', 103, 90),  
(5, 'Neha', 102, 60),  
(6, 'Rohan', 103, 75);
```

a)

```
CREATE TABLE STUDENT (  
    Student_ID INT PRIMARY KEY,  
    Student_Name VARCHAR(50),  
    Course_ID INT,  
    Marks INT  
);
```

b)

```
SELECT *  
FROM STUDENT  
WHERE Marks > 70;
```

c)

```
SELECT S.Student_Name, C.Course_Name  
FROM STUDENT S  
JOIN COURSE C  
ON S.Course_ID = C.Course_ID;
```

d)

```
SELECT Course_ID, AVG(Marks) AS Average_Marks  
FROM STUDENT  
GROUP BY Course_ID;
```

e)

```
SELECT Course_ID, AVG(Marks) AS Average_Marks  
FROM STUDENT  
GROUP BY Course_ID  
HAVING AVG(Marks) > 65;
```

Q2.

Create EMPLOYEE table

```
CREATE TABLE EMPLOYEE (  
    Emp_ID INT PRIMARY KEY,  
    Emp_Name VARCHAR(50),  
    Department VARCHAR(50),  
    Salary DECIMAL(10,2),  
    Joining_Date DATE  
);
```

Insert data

```
INSERT INTO EMPLOYEE VALUES  
(1, 'Amit', 'IT', 60000, '2023-01-10'),  
(2, 'Neha', 'HR', 45000, '2022-05-15'),  
(3, 'Rahul', 'IT', 75000, '2021-08-20'),  
(4, 'Priya', 'Finance', 55000, '2023-03-12'),  
(5, 'Rohan', 'IT', 65000, '2020-11-05'),  
(6, 'Sneha', 'HR', 48000, '2024-01-18'),  
(7, 'Karan', 'IT', 80000, '2019-07-25'),  
(8, 'Pooja', 'Finance', 58000, '2022-09-30');
```

a)

```
SELECT *  
FROM EMPLOYEE  
WHERE Department = 'IT'  
ORDER BY Salary DESC;
```

b)

```
SELECT MAX(Salary) AS Highest_Salary  
FROM EMPLOYEE;
```

c)

```
SELECT Department, COUNT(*) AS Employee_Count  
FROM EMPLOYEE  
GROUP BY Department  
HAVING COUNT(*) > 3;
```

d)

```
SELECT *  
FROM EMPLOYEE  
WHERE Salary > (  
    SELECT AVG(Salary)
```

```
FROM EMPLOYEE  
);
```

e)

```
CREATE VIEW IT_Employees AS  
SELECT *  
FROM EMPLOYEE  
WHERE Department = 'IT';
```